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Metabolic thinking in planning and designing urban landscapes: practitioners' perspectives on agency

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ABSTRACT

Urban metabolism approaches are increasingly becoming relevant in providing pathways to sustainable development. The ways through which urban green infrastructures are planned and designed can influence resource flows in cities. Yet, little is known about the agency and contributions of landscape architects and spatial planners as curators of urban greenery, in facilitating circular resource flows, and accelerating the transition to a circular economy. This paper offers an empirical inquiry into the understanding and use of metabolic approaches in spatial planning and design practices. Through a multi-method approach and the use of boundary spanning as an analytical lens, we identify different modes of agency of spatial practitioners in enhancing resource efficiency across the life stages of urban landscape projects. Results show that while the 'urban metabolism' concept is not widely used by practitioners, its relevance to planning and designing green infrastructure for sustainable resource flows is well understood. Additionally, we highlight existing barriers including policy constraints in engaging with metabolic approaches and offer recommendations to facilitate a resource-sensitive turn in planning and designing urban landscapes. This study highlights important contributions of design practice to an expanding discourse of urban metabolism, including policy reforms for enabling circular resource flows in cities.

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1. Introduction

With the current rates of land, resource and biodiversity depletion, how we design and build our cities needs to change if we want to avoid breaching the earth's planetary boundaries. Cities are responsible for 56–78% of all anthropogenic energy use (Grubler et al. 2012) and associated greenhouse gas emissions (UN-HABITAT 2022). Cities are also driving material extraction with more than 50% of all material stocks used in the built environment (Krausmann et al. 2017). Understanding urban resource

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and energy flows is key to realizing what it will take to promote a transition from resource-intensive built environments towards alternatives that manage resources more carefully (United Nations Environment Programme 2018). The progressive acknowledgement of urban environmental problems and their impacts on climate change has turned the spotlight on the study of the 'Urban Metabolism' as an operational base for sustainable urban development (Pistoni and Bonin 2017).

Urban Metabolism (UM) is an interdisciplinary concept that '*analogizes urban systems to living organisms and adopts the metabolic thinking of living organisms to urban operations because they consume resources from the environment and excrete waste*' (Cui 2018, 704). The study of UM is used to understand how cities extract, manage and utilize resources and emit pollution and waste (Lanau, Mao, and Liu 2021), by characterizing the social-technical-ecological processes around material flows (Kennedy, Cuddihy, and Engel-Yan 2007). This is a widely used definition built upon the Industrial Ecology conceptualization of the term. In this paper we use the term '*metabolic thinking*', not as a metaphor, but rather a *modus operandi* for simply thinking in resource flows and their metabolic independencies (socio-ecological processes of circulation and transformation (Ibanez 2021)), as a way forward in applying resource sensitive design and spatial planning strategies across urban scales (Perrotti and Iuorio 2019; Perrotti, Moosavi, and Otero Peña 2023). We argue that the integration of metabolic thinking in urban landscape design and planning has not been fully mobilized in practices, and the agency of spatial practitioners in this integration is understudied.

The aim of this paper is to unpack and map the perceived modes of agency and contributions of urban landscape designers and planners in employing metabolic thinking, and to understand the barriers and facilitators of resource-sensitive design – to sustainably design with resources using landscape-led spatial configurations as levers.

To address the aim of this research, we used an empirical inquiry into the perspectives of spatial practitioners focusing on the following key question:

> *How do practitioners employ metabolic thinking in design and planning processes for urban landscapes?*

The key question is addressed through an empirical inquiry into the agency of design practitioners in making resource-related decisions throughout the life stages of the production of built environments. The novelty of the approach lies in using mixed methods and triangulation of findings, across a survey, focus group workshop and in-depth interviews (outlined in Figure 1). This approach allowed us to gain an overview of how spatial practitioners conceptualize and understand the urban metabolism concept, but also an in-depth investigation of the ways through which they engage with its underpinning principles in design, and the challenges they face. We first provide a review of the existing literature on the integration of green infrastructure (GI) in UM studies, and the ways through which the agency of spatial practitioners are understood in these studies.

2. Background

2.1. Operationalizing the concept of urban metabolism in planning and design

The multiplicity of disciplinary perspectives on the UM concept has created a conceptual plurality around the term, making it difficult to imply a universal approach to UM (Bahers et al. 2022).

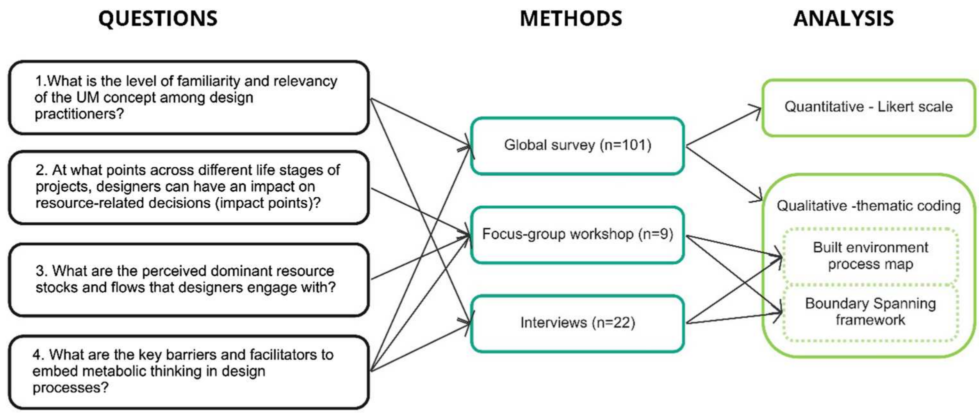

Figure 1. Research design.

Beyond the concept, UM research provides a variety of approaches, analytical tools and methods to assess the resource intensity of urban systems (Perrotti 2020). One of the most commonly employed UM models is material flow analysis (MFA), which allows the classification and quantification of material flows within the boundary of a defined system (Brunner and Rechberger 2016). Other common UM models include life cycle assessment (LCA), ecological network analysis, emergy analysis and ecological footprint (EF) among others (Zhang, Yang, and Yu 2015).

While the importance of understanding UM is increasingly recognized for reflecting urban complexities, its projective dimensions and practical applications in planning and designing urban environments (Ibanez 2021), particularly urban GI remain few (Perrotti 2020). Given the multiplicity of disciplinary perspectives on UM, Broto, Allen, and Rapoport (2012) highlight the need for interdisciplinary dialogues to develop both theoretical and practical approaches to UM, and derive ideas that can be made operative in planning and design processes. Despite a number of studies that aim to operationalize UM approaches in urban planning and design (Kennedy, Pincetl, and Bunje 2011), the uptake of metabolic accounting methods and models by urban landscape designers has been slow (Pasini 2022).

This could be due to multiple reservations specific to the limitations of UM accounting models including: (1) the failure of common accounting models such as material flow analysis, ecological footprint and life cycle assessment in capturing the spatial qualities of landscapes and territories (Bahers et al. 2022; Galan and Perrotti 2019; Otero Peña, Perrotti, and Mohareb 2022); (2) their limitation in accounting for various positive benefits such as impact offsetting that ecosystems can bring to urban flows (Birkeland 2012; Cárdenas-Mamani and Perrotti 2022) and (3) the complexities around capturing temporal and dynamic qualities of GI and assessing their long-term performance (Larrey-Lassalle et al. 2022). Furthermore, as detailed in (Perrotti 2019), limited specialist knowledge, silo-thinking and the lack of a common language among relevant disciplines can limit the effectiveness of science–practice knowledge transfer in applying UM assessment in urban design and planning.

These challenges and the multiplicity of perspectives highlight the concept of UM as a *boundary concept*, intersecting different knowledge areas and crossing disciplinary boundaries. 'Boundary work' theory has been useful in addressing the complexity inherent in natural resource management through inter- and transdisciplinary communication, translation and mediation (Mollinga 2010; Nassauer 2023). It offers strategies to support the decision-making of actors by reconfiguring the boundaries between different forms of knowledge, academic and non-academic expertise, and interests and values (Opdam et al. 2015). In crossing disciplinary boundaries and science-policy-action integration, the critical role of 'boundary spanners' has been highlighted by a number of scholars (Schröter et al. 2023; van den Brink et al. 2019). We adopt boundary spanning as a lens to investigate the agency and contributions that spatial practitioners can bring to operationalizing the UM concept in urban landscape planning and design.

2.2. The relevance of metabolic thinking in planning and designing urban landscapes

The underlying territorial landscape and the urban fabric highly influence urban metabolic flows (Thomson and Newman 2018), highlighting the need for spatially explicit approaches to UM (Bahers et al. 2022; Kennedy, Pincetl, and Bunje 2011; Perrotti and Stremke 2020). Studies have shed light on the provided ecosystem services by GI and the impacts on the outcomes of material and energy flow accounting in urban areas (Cárdenas-Mamani and Perrotti 2022). Complementarily, two decades ago, Mathew Gandy shed light on the importance of analysing the metabolism of urban spaces and landscapes to illustrate the transformation and redefinition of natural ecosystems into the urban fabric and built environments (Gandy 2003). Following this line of thought, other scholars have emphasized how UM assessments can improve the projective potentials and environmental and spatial qualities of new designed metabolic landscapes such as wastescapes, energyscapes and waterscapes (Amenta and Van Timmeren 2018; De Meulder, Marin, and Shannon 2022; Oudes and Stremke 2018).

The employment of environmental impact assessment tools and accounting models such as MFA and LCA beyond buildings to assess the direct and indirect environmental on-site and off-site impacts of GI projects is an emerging area of scholarship (Babí Almenar et al. 2023; Romanovska, Osmond, and Oldfield 2023; Stephan et al. 2022). These and other studies highlighting the need for urban landscape designers to more systematically engage with environmental performance of sites across their life cycles (Moosavi, Stephan, and O'Dea 2022).

2.3. The agency of spatial practitioners in urban metabolism

Recent research points to the need for deeper understanding of human and non-human agency as an essential step to fully capture the dynamics of resource circulations underpinning GI and the built environment at large (Perrotti 2022). Within the scope of this research, we distinguish 'agency' from 'role' in that, through agency, transformative change may occur in a socio-ecological system, beyond normative actions, toward actions that transform practices. This framing resonates with O'Brien's definition stating that '*activating conscious human agency that is critically reflective of individual and shared assumptions, beliefs and paradigms is a powerful way to shift norms and institutions in ways that support [transformation]*' (O'Brien 2018, 159).

In relation to UM studies, the concepts of 'agency' (of both humans and nature) and 'UM' have been mobilized in urban political ecology since its inception (Zimmer 2010) as a way of framing the relations societies establish with nature and the circulations of resources as political and socio-ecological processes (Gandy 2018). In parallel, more-than-human approaches and the application of Actor-Network-Theory (Latour 2007) and of agentic readings of urban assemblages in urban studies (Brenner, Madden, and Wachsmuth 2011; Färber 2019) have raised fundamental questions of agency for UM research: *who/what* has the capacity to affect resource access and distribution in an urban assemblage, under what conditions (*where/when*), and through which *modus operandi* (*how/to what extent*) (Bertoldi and Perrotti 2025).

Despite an emerging body of literature on the potential of landscape theory and practice to prompt new ways of knowing and investigating UM (Marin and De Meulder 2018; Perrotti 2019, 2022; Pistoni and Bonin 2017), the agency of spatial practitioners as key decision makers in shaping metabolic processes is understudied, and this study builds on an emerging body of works that aim to address this gap through taking an empirical multi-method investigation, detailed below.

3. Method

To address the aim of this research, a multi-method approach was used. First, we used an international survey to gather how urban landscape designers/planners understand the UM concept, and perceive the relevance of metabolic thinking in design practice, globally. It also aimed at identifying the key barriers and limitations to engage with resource efficiency through metabolic thinking. This quantitative approach was complemented with qualitative data collected via a focus group workshop with invited spatial practitioners, mainly practicing in Belgium. The workshop further investigated the agency and contributions of designers/planners in mobilizing metabolic thinking across the life stages of projects and the challenges designers face. Additionally, for an in-depth understanding of the barriers and facilitators to resource-sensitive design approaches, we conducted 22 interviews with design practitioners in Belgium. Figure 1 depicts the research design approach, including the analysis methods elaborated in section 3.4.

3.1. Survey

An online survey was designed and distributed globally (between 18th of May and 30th of November 2022), using the Qualtrics survey software. Questions included quantitative 5-point Likert scale, preference, multiple-choice and rating questions along with qualitative open-ended questions (see Supplementary Materials for the list of questions). In addition to questions capturing the disciplinary background of professionals, location and size of practices, the questionnaire was designed around three major themes related to UM: (1) familiarity, (2) relevance and (3) key challenges or barriers. The survey was distributed via different platforms, including social media, personal emails to the authors' networks of professionals, and snowballing. The International Federation of Landscape Architects in Asia, Europe and Africa also distributed the survey to members via their newsletter. In total 101 individuals responded to the survey ($N = 101$).

3.2. Focus group workshop

A half-day workshop was held in Brussels, Belgium on 30th of June 2022, with invited practitioners mainly based in Belgium. To identify potential participants, first the list of registered landscape architects from the official website of *The Belgian Association of Landscape Architects* (ABAJAP-BVTL) was consulted. It is important to note that in Belgium the 'landscape architecture' profession is not yet considered an accredited titled profession, and many licensed architects and urban planners and designers that work at the urban landscape scale are registered under the national landscape architecture association because of the type of projects they work on. This can be related to limited education programmes across Belgium offering a master's degree in landscape architecture (i.e. only one master's programme) (IFLA Europe 2022).

We limited our selection to practices with a public profile presence and those involved in urban landscape projects. In total, 21 practitioners responded to the email invitation (out of 78 contacted). Eventually, 9 practitioners were available to participate in the half-day workshop. The participants had different disciplinary backgrounds (e.g. urban design, spatial planning, landscape architecture, sustainability), resulting in a relatively heterogeneous sample.

3.3. Interviews

Semi-structured interviews with 22 design practitioners were conducted between 27th of February and 10th of October 2023. Participants were identified and shortlisted through a similar approach detailed in 3.1.2, with a total of 52 practices invited for the interview. In total 22 responded to the invitation. Interviews were conducted either in person, or online via Zoom and took 50–60 min. Consent was sought in line with human ethics protocols. Given the purpose of the interviews was to explore an in-depth understanding of the barriers and facilitators in engaging with metabolic thinking, the findings are not intended to be statistically representative of Belgian design practices. However, the captured viewpoints through the interviews and workshop *do* provide insights to the ways the professionals engage with resources and the challenges that they currently face.

3.4. Analytical framework

As depicted in Figure 1, we used a mixed-method approach for the analysis of data. The survey data were analysed and presented quantitatively. The recordings of the workshop and interviews were transcribed and qualitative analysis with thematic coding was undertaken by the lead author in NVivo, and cross-checked with the author team. Initial coding was based on key questions around (a) impact points across projects' life stages, (b) dominant metabolic flows relevant to the practitioners and (c) barriers and facilitators of resource sensitive approaches. We used the built environment process map developed by Hürlimann et al. (2021) to map the impact points and barriers/facilitators across different life stages including: (1) Change Initiation; (2) Strategic Planning; (3) Project Consideration; (4) Design; (5) Costing and Approvals; (6) Construction; (7) Use/Management; and (8) Renewal/Recovery/Decommissioning. We then categorized the responses across four boundary spanning roles of identified in van den Brink, van den Brink, and Bruns (2022), including: (1) the subject-based designer, (2) the visionary narrator, (3) the process-based designer and (4) the design-led entrepreneur. This framework proved relevant to understand the impact, agency and contributions of spatial practitioners in engaging with theoretical and practical underpinnings of the concept of UM.

4. Results

The survey captured 101 responses in total. The first part of the survey aimed at capturing background information about the practitioners' expertise and their practice. Among the respondents, the majority were landscape architects (64%) and urban designers (32%), followed by architects (18%) and urban planners (16%). The majority (47%) of survey respondents described their practice as being small (less than 10 members), followed by medium (31%), and large practices (22%). In terms of geographical distribution, we collected responses from 37 countries, from all continents, with a higher response rate from countries such as Australia, Belgium and Netherlands, reflecting where the networks of the authors were mainly established.

As mentioned above, the survey was complemented by a workshop and 22 interviews. Table 1 captures the expertise and sectors of the focus group workshop participants, with 4 out of 9 as landscape architects. It also includes the codes used in the analysis and reporting. Table 2 shows the breakdown of interviewees in terms of their region, sector and firm size. We aimed for a balance between practitioners working in Wallonia, Flanders and Brussels regions relevant to the population of registered designers in each region. Most interviewees worked in small private practices, and all but two had a landscape architecture education background.

4.1. The familiarity and relevancy of the UM concept to design practice

The results of the survey showed that there is a general agreement on the relevance of UM in designing urban open spaces with 78% of respondents indicating extreme (19%) to very high (59%) relevance, with a mean of 2.1, median of 2 and a Standard Deviation of 0.8 on a 5-point Likert scale (1 = extremely relevant, 5 = not relevant at all), indicating a relative agreement. However, the overall level of familiarity of UM and its key principles and approaches among the surveyed practitioners is low, with half of the respondents (50%) selecting slightly familiar (31%) or not familiar at all (19%), with a mean of 3.56, median of 3 and a Standard Deviation of 1.05 on a 5-point Likert scale (1 = extremely familiar, 5 = not familiar at all), indicating a significant spread of responses. Among the other half, only 19% of respondents indicated they were familiar with the UM concept and its main approaches (see Figure 2 and 3). This was also reflected in responses to a

Table 1. Expertise of workshop participants and their codes.

Codes	Expertise	Sector type
P1	Urban Planning & Sustainable Development	Private
P2	Architecture and Urban Design	Private
P3	Landscape Architecture	Private
P4	Landscape Architecture	Private
P5	Architecture, Urban Design, Spatial Planning	Private
P6	Landscape Architecture	Public
P7	Landscape Architecture	Public
P8	Urbanist/Architecture	Private
P9	Urbanist/Architecture	Private

Table 2. Breakdown of the interviewees' profiles.

	Region			Sector type		Office size		
Characteristics Codes I = Interviewee	Wallonia (French) I6; 10; 11; 13	Flanders (Dutch) I2; 3; 5; 9; 12; 16; 17; 18; 19	Brussels (both) I1; 4; 7; 8; 14; 15; 20; 21; 22	Public I12; 20; 22	Private All others	S	M	L
Number of Respondents	**4**	**9**	**8**	**3**	**18**	**11**	**5**	**6**
Total interviews	**22**							

multiple-choice question on the definition of UM, where 18% indicated they are not familiar with the UM concept. One survey respondent indicated that:

> *Urban metabolism is a concept that I'm not familiar with; prior to filling this form I had not heard of this term but from its definition and due to my experience working on various projects that have required some level of energy management and sustainability, I have a good understanding of what it requires and what the potential benefits are.*

In parallel to the results of the survey, the discussions in the workshop revealed that an understanding of the existing flows in landscapes is highly relevant to design decision-making that incorporate 'systems thinking', albeit not being framed as UM:

> *we are one of the few professions that is able to translate 'systemic thinking' to space, making 'concrete' the sometimes 'vague' discourses of urban metabolism, circular economy, etc. (P5)*

The survey results also revealed that UM approaches are understood to be more relevant for decisions at the 'regional scale' (41% of respondents), with an even spread across all the other scales (metropolitan, district/neighborhood, site). The influence of 'scale' also emerged in discussions during the workshop, as an important factor determining the impact that designers have on resource-related decisions (see section 4.2).

Interview results also confirmed the low level of familiarity with the concept of UM in how the respondents reacted to the question '*how do you define metabolic thinking?*' Most respondents could not immediately think of a definition.

> *I don't really know what you mean. So I don't know what it is. I think that's maybe a bit off from what we do (I22)*

In these cases, the researchers provided a widely accepted definition, as an *approach to understand how cities extract, manage and utilize resources and emit pollution and waste*. However, when a definition of UM was provided by the interviewer, the majority of respondents expressed that they engage with these approaches and to them, it is about thinking in *systems* that are *interconnected* and enabling *natural processes* to occur, which landscape architects often do.

> For me, metabolic thinking is really incorporated in what good landscape architecture does, and it is taking into account, let's say, the construction of the lands from source to sink, so metabolic thinking is incorporated in any good systemically designed landscape. But I'm not sure if the term would resonate with many practitioners. (I18)

The language that practitioners used in the interviews to conceptualize 'metabolism' revealed an array of terms and concepts, some related to the theoretical underpinnings of the UM concept ('*what*'), while others focused on action-oriented framings ('*how*').

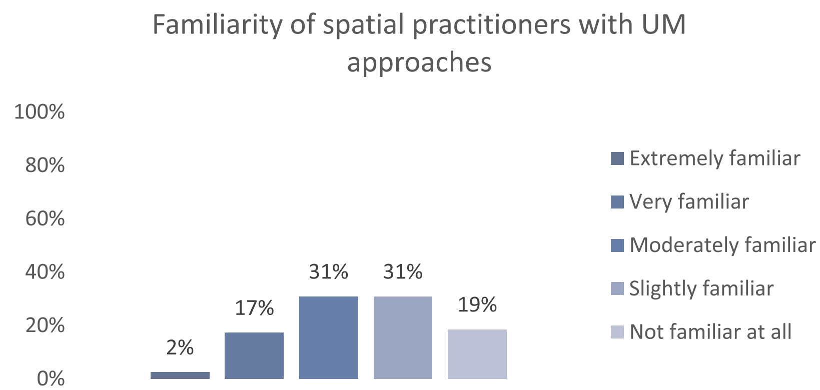

Figure 2. Survey results showing the percentage of familiarity with UM approaches.

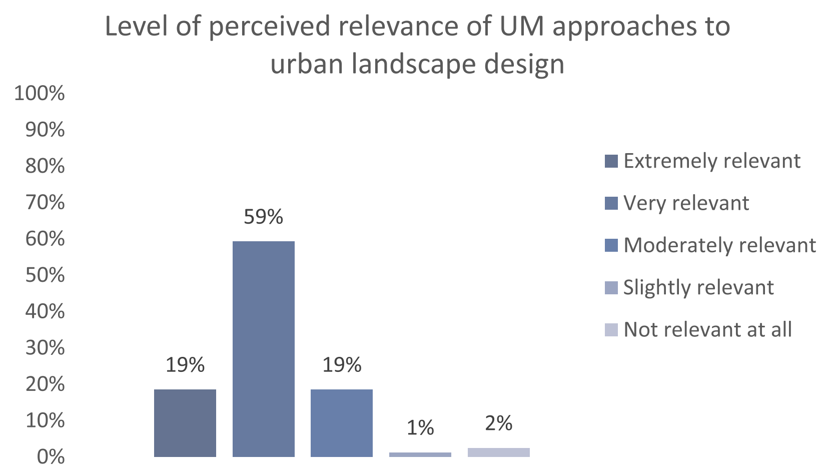

Figure 3. Level of perceived relevance of UM approaches by spatial practitioners.

Through thematic coding of all responses to this question (included in Supplementary Materials) we synthesized these terms as presented in Table 3. The action-oriented framing was further categorized into those that inform the analysis of sites (e.g. understanding the history of sites), and those that are projective and inform design interventions (e.g. reuse of site materials). This expanded conceptualization of UM highlights the synergies among interdisciplinary knowledge areas that designers can lever to embed metabolic thinking in design workflows, and contributes to establishing a practice-specific conceptualisation of UM.

4.2. Levels of engagement with metabolic thinking across life stages of projects – identifying impact points

To understand the agency of designers in integrating metabolic thinking in design processes, we asked the participants of the workshop to map the key impact points where designers can integrate metabolic thinking in influencing resource-related decisions across the built environment process map (by Hürlimann et al. (2021)) (see Figure S

Table 3. Key concepts related to urban metabolism and terms mentioned by the interviewees.

	Key identified themes	**UM related terms mentioned**	
Theoretical underpinning	Systems thinking	Thinking in systems, Systemic design (I8; I19) Landscapes as interconnected systems (I14) Systematically designed landscapes (I18) Design with living systems (I2) Cascading ecosystem services (I8)	
	Circular economy	Circular thinking (I9) Source to sink (I18) A full cycle (I21)	
	Landscape Urbanism	Regenerative development, Re-use of post-industrial sites (I3) Blue and green landscapes as backbones of urbanism (I5) Landscape as transect (I4) Deep understanding of ecological design (I12)	
Action-oriented framing	Sites as multilayered palimpsests	Working with the history of sites (I4) Understanding existing flows (I20) Preserving nature/natural processes (I7)	*Analytical*
	Resource use/cycles	Optimize flow of nutrients (I11) Reuse of space, Reuse of site materials, waste as materials Living (biobased) materials/soils (I1; I10) Avoiding over designing (I20)	*Projective*
	Resource production	Energy landscapes, Food production (I9; I16)	*Projective*
	Lived experiences	Inclusive places (I10) Places of experiences (I2) Material world and lived experiences(I2)	*Analytical/ Projective*

in Supplementary Materials). To facilitate this, the participants were divided in smaller groups of 3.

Most participants agreed that the analysis, infrastructure planning, design and construction phases were critical for them, as designers, to embed metabolic thinking in the process. However, there were some disparities regarding whether designers have greater impact at the early stages of the project compared to later stages where projects will be realized and built. Three important aspects related to the shifting impact of designers across the process were discussed: the 'scale' of projects, how the 'brief or tenders' were defined, and the 'policies and regulations' affecting the design and construction.

Those participants who often worked at the large scale argued that they had greater impact at the early stages of design, with '*setting the tone*' (P2) and ambitions for more sustainable outcomes. This was not the case with practitioners who worked on smaller scale projects, often for private clients. They suggested UM approaches tend to have greater impact in the construction phase, through material and plant choices and specifications, and negotiating with contractors. Participants argued that contractors with a business-as-usual mindset often follow rigid protocols which leads to discrepancies between design documents and the built outcome, and to change that is challenging:

> *It is often the normative framework that hinders changes even if you have intensions to include metabolic thinking, it is so rigid that sometimes it is really difficult to change the system* (P6)

It was discussed that designers' engagement with UM approaches should ideally have an impact from the early stages all the way to the construction phase, consistently influencing the process and outcome. This view is reinforced by this statement:

> *It is always a big effort depending on the process and who you are involved with – but it's important that everyone plays the same card, and that's sustainability, and maintaining the efforts throughout the whole process down to the contractors involvement* (P3)

A number of interviewees mentioned the challenge of having impact at the early stages of change initiation, because landscape architects and urban designers are often brought into projects very late, when decisions with greater resource impacts are already made.

From the discussions in the workshop and interviews we observed that the level of impact across the life stages of projects is also partly determined by the characteristics of design firms. We used the four boundary spanning roles as a matrix to map the typologies of practices. This allowed us to link the responses around impact points to the boundary spanning framework. For example, those practitioners that take the role of the subject-based designer and focus on small scale projects for private clients, considered their influence as being greater at the design to construction stages, compared to those practices that consider themselves as process-based designers and work on larger scale projects, who believed they have equal impact across all life stages. Figure 5 presents the level of impact across project life stages across the four boundary spanning typologies (see section 5.1).

4.3. Dominant metabolic stocks and flows perceived by practitioners

The term *stock* – referring to material stocks accumulating in society including buildings, infrastructures, machinery and other long-lived products (Streeck et al. 2020) – was not explicitly mentioned by any of the participants, but they referred to soil, plants (biomass) and in-use materials, which are defined as stocks in resource accounting UM approaches.

In contrast, a *flow* is a variable measured over time, tracing the required process for production, processing, use and management of waste. In addition to standard flows considered in metabolic analysis (e.g. energy, greenhouse gas emissions, waste), the designers perceived other flows including water (stormwater runoff), mobility and people. When asked about the flows they dominantly engage with, some practitioners talked about 'services or functions', such as climatic conditions, biodiversity and ecosystem services more broadly.

One theme that frequently surfaced in the focus group and interviews was the role of designers in engaging with nonnumeric flows related to socio-cultural dimensions. An in-depth understanding of the community's socio-cultural practices was discussed in relation to two different dimensions: (a) how the community uses a particular site and its natural resources, and what sort of social behaviours they practice and (b) dominant placed-based behavioural patterns of consumption, particularly in terms of energy use, waste management or mobility.

Some examples were shared, including cultural practices among religious communities and the preference for gas top cooking, and how it influences infrastructure and utility provisions, or energy transitions (P5). Another example was given on urban design interventions to address how communities deal with extreme heat in low-income communities (P3). For example, if the majority of the community live in small apartments with no garden or terraces and find coping mechanisms such as drinking more water or staying in shades of trees in public spaces, the urgency of the solutions should be directed at improving public facilities such as parks with large tree canopies. It was highlighted that understanding behaviours and practices linked to resource use, enables designers to define the urgency and suitability of interventions and programmes, which can then influence how resources are managed and circulated.

In parallel, the analysis of interviews revealed the increasingly important role that designers can play in resource-related decisions beyond physical interventions to sites, with a strong focus on changing social norms in everyday practices. One interviewee discussed a project that focused on nature-based decarbonization and catalysing sustainable farming and changes in dietary behaviours to leverage net zero land management at the regional scale (I8). Other examples of engaging with 'practices' were shared focusing on smaller scale projects, including the role that designers can play in changing the assumption of considering private landscapes '*as a luxury*' (I9), and shifting the mindset of clients that systematically ask for swimming pools in their backyard (I19; 22), which dismisses opportunities for enhanced biodiversity and reducing ecological impact (I17).

Table A in the Appendix captures a list of flows, stocks and practices that were discussed by workshop participants and interviewees.

4.4. Perceived key barriers and facilitators to embedding metabolic thinking in design practice

We identified the top key barriers in engaging with UM approaches from the perspective of practitioners through a ranking question in the survey (see Figure 4). The 'lack of understanding of basic metabolic processes or dynamics' was selected the most in the top 5 barriers (33% across all rankings) and was ranked 1st by the majority. Other top selected barriers were 'lack of data' (26%) and 'limited competencies for engaging with quantitative data' (25%). Top-down governance systems for resource management (24%) and dominating engineering approaches were also selected as important barriers (24%).

During the workshop, an exercise was designed to facilitate discussions, with the circulation of design documents of an existing landscape project (a built community park near), asking the groups to reflect on barriers, data needs, tools and skillsets required to re-design the park considering multiple metabolic flows. Through the interviews we further examined the barriers in more depth by asking the practitioner to expand on the top three barriers they face in thinking through resource flows in projects. Five main categories of barriers were identified across all data: (1) knowledge barriers (informational-technical-technological), (2) governance barriers, (3) financial barriers, (4) socio-cultural barriers and (5) physical-environmental barriers. Sub-categories of barriers were also identified across all responses (workshop and interviews). Table 4 provides a summary of key barriers and facilitators that emerged in this study under each theme.

While knowledge-relevant barriers were seen as a critical barrier in the survey and among the workshop participants, the discussions in the interviews mostly centred around lack of finance and restricted timelines in projects. This was perceived as an impediment for deep engagement with data collection, understanding metabolic flows in relation to the site, and beyond site boundaries.

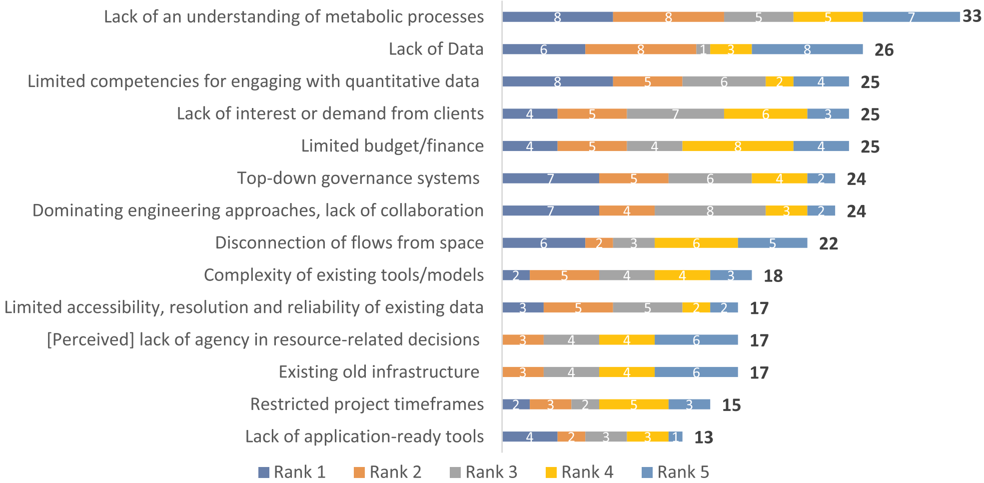

Figure 4. Existing key barriers in engaging with UM approaches in practice (by numbers of survey participants' ranking them in each of the top five ranks).

Some argued that it is a question of redefining project fees, and reforming what activities are paid in commissions. The system where design fees are a percentage of total construction fees was particularly criticized, as it encourages some designers to use costly materials such as concrete with high associated emissions, just to increase their fees (I14). Some suggestions were offered such as assigning values to process-oriented designs and research activities to incentivize more ecological design approaches underpinned by metabolic thinking.

In working with large scale projects, the lack of methods for measuring and assessing design decisions from a monetary perspective was mentioned. One interviewee highlighted the difficulty of translating multiple layers of data on flows into a 'tool' that is simple and accessible to the community of design practice (I15). Others mentioned the disconnection of UM accounting tools from design modelling tools and the lack of training opportunities for designers. More room and financial means for experimental approaches were seen as enablers to address these challenges.

> *It would be nice if there was general funding for this kind of experimentation that doesn't put reliability on the designer (P7)*

It was also argued that understanding the site with a metabolic lens requires delving into quantitative and qualitative data on different flows, which requires time to collect data that are often not readily available. Participants argued that this 'time' is not included in project briefs, or commission fees.

> *Balancing the pace of academia and practice, practice is fast, to align this timing is a challenge (P5)*

Values underpinning projects tenders and briefs as well as clients' sustainability ambitions were identified as critical barriers by most interviewees. Particularly, in working with private clients, it was highlighted that shifting people's mindsets is difficult.

Table 4. Perceived dominant barriers and mentioned facilitators in engaging with metabolic thinking in practice based on interviews and workshop data.

Themes of barriers	Perceived barriers or challenges	Mentioned facilitators
Physical-environmental	Links between space and resource flows Thresholds between ecosystem services and utility services Ecosystem services conflicting with mobility services Competition for space (cars vs GI) Tensions btw durability and sustainability	Balancing environmental values of spaces with socio-economic benefits Mobilizing temporary and mixed-use zoning through strategic planning
Knowledge (Informational-technical)	Lack of data or dispersed databases Lack of training Lack of knowledge and confidence in recycled materials Tabula-rasa approaches to landscapes Balance between practice and caring for the environment Disconnection of UM tools with modelling tools	Centralizing information Ease of access to information Integrating tools with design modelling platforms Room and time for experimentation Training for spatial practitioners
Socio-cultural	Working with communities/clients with rigid mindsets Mindsets of contractors Disparate values/ priorities (Unsustainable) lifestyles	Communicating the benefits e.g. use of value boards Engagement with the community through site visits and visual communication Participatory approaches
Governance	Business-as-usual approaches Rigid protocols and automatism Safety regulations Normative frameworks-pre-defined templates Lack of sustainability criteria and standards in tenders Short-termism – rapid political cycles	Standards for minimum use of recycled materials Inclusion of performance criteria and indicators in tenders or briefs Sustainability ambitions embedded in project briefs
Finance	Financial restrictions in construction budget High construction fees for sustainable materials No commission fees for research and testing Prioritizing budget for buildings not landscapes	Incentives for using local recycled materials Introducing ecological tax Funds for experimentation Redefining commission fees

> *Creating a cultural shift where people realize that a certain style of gardens and public space that is very material based is not the right thing to do and that a more natural approach is where you just do nothing in some parts and you leave more space for natural processes (I17)*

There were diverging opinions around whether it is the responsibility of the designer to invest themselves in going beyond the brief and usual responsibilities, and make extra effort to undertake research;'acting as an environmental activist' rather than as a conforming consultant. One designer questioned the role of the designer as mere consultants:

> *[…] but it comes down to you, as designers, how much your work limits itself to developing a plan, or should it be about taking care of the environment and curating it with care (P2)*

In contrast, some designers argued that eventually design is a 'commercial activity', with one participant stating:

> *You wouldn't do it just because you care. You need to earn money; at some point you need to balance means for in-depth study and the practice running. It is a very delicate balance to find! (P6)*

This resonates with the concept of crossing boundaries or boundary spanning beyond what is expected from design consultants. We now discuss the findings against the boundary spanning typologies.

5. Discussion

5.1. Mapping the agency of designers in urban metabolism approaches

From the multilayered analysis of data across the survey and workshop, we conclude that the agency of designers and spatial planners in engaging with metabolic approaches in urban landscape projects takes different forms depending on several factors, including the type of commission, scale of projects and the profiles of practices involved. This is line with the findings of Pistoni and Bonin (2017) focused on practices in Rotterdam and Amsterdam, where scale of projects determined the engagement with metabolic approaches.

One key theme that emerged was the tension between perceiving design practice as a commercial activity, or as a research-through-design activity to catalyse change. This has been a concurrent focus of a number of scholarly debates around the relationships between design and research, or between intuitive and scientific inquiry (Cortesão and Lenzholzer 2022; Kwak, Deal, and Mosey 2021; Moosavi 2022).

While there is general awareness among the community of practice in urban landscape design and planning about the need for understanding systems of resource stocks and flows in leveraging resource-sensitive designs, metabolic accounting methods, tools and models are not yet mainstreamed in design and planning practices. Nevertheless, the synthesis of our findings show that designers do engage with metabolic thinking in different ways, particularly through multiscalar systems thinking; even though these approaches are not explicitly framed as urban metabolism approaches among the community of practice. These 'different ways' of engaging with metabolic thinking are mapped against the quadrant matrix of the boundary spanning roles of designers presented in Figure 5 and discussed below.

It is important to note that spatial practitioners often do not neatly fit within one of these four profiles, and depending on the type and scale of commissions, they may change roles across these categories.

1. *Subject-based designers* often work on plot scale projects and can have an impact through understanding the existing stock-flow relationships, and selecting and translating those data into designs with low environmental impacts (for example by advocating the use of recycled local materials). They may have to overcome challenges such as working around existing utility infrastructure and rigid regulations (McLean and Roggema 2019), competition for more space for urban nature-based solutions (Moosavi, Browne, and Bush 2021), and managing conflicts with contractors with often rigid mindsets to ensure there are no discrepancies between what they design and what is executed (Moosavi et al. 2023). To increase their agency, they need to

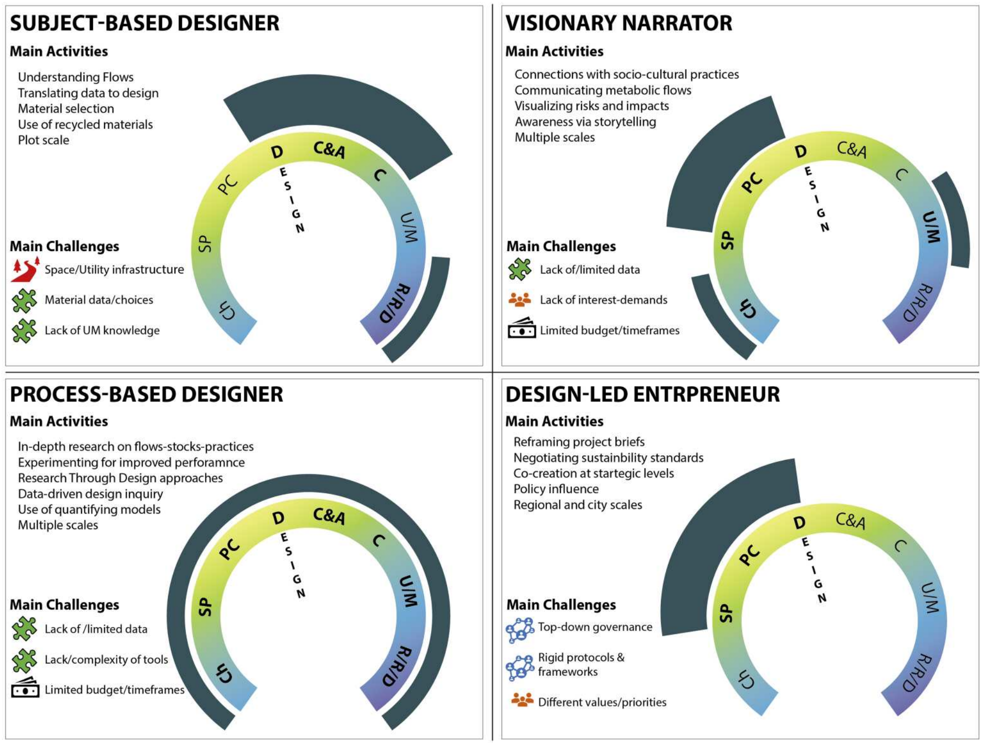

Figure 5. Mapping the perceived agency of designers in engaging with metabolic thinking against a quadrant matrix of boundary spanning roles (adapted from van den Brink, van den Brink, and Bruns (2022)), showing the different impact points across life stages of projects (thicker arcs = greater impact, thinner arcs = lower impact) Project life stages: Ch = Change Initiation; SP = Strategic Planning; PC = Project Consideration; D = Design; C&A = Costing and Approvals; C = Construction; U/M = Use/Management; R/R/D: Renewal/Recovery/Decommissioning.

expand their understanding of existing flows and their dynamics over time (referred to as fluxes here) beyond the physical boundaries of the site. They also need to have great negotiation skills to convince the clients, communities and contractors for sustainability agendas.

2. *Visionary narrators* can have an impact at setting the ground and visions for project initiation and strategic directions in early stages of projects. They can build an understanding of on-going patterns of resource consumption and elimination of waste through speculative narratives. Having strong communication and storytelling skills, their agency lies in impactful visualization of metabolic (visible and invisible) flows, and projecting and speculating possible scenarios to create awareness. They can link existing socio-cultural practices with current and future patterns of metabolic resource flows, and to ground them in the spatiality of places. Through directly working with communities with diverse socio-cultural values, the challenges they face often relate to the lack of awareness, demand, or interest from clients. This is also considered a key barrier by Amenta and Van Timmeren (2018). These may be related to disparities of values and priorities, or the lack of appetite for sustainable outcomes. While their contributions appear to remain speculative and emancipatory,

they have a far-reaching influence by creating an appetite for solutions that contribute to a more circular metabolism.

3. *Process-based designers* often act as coordinators connecting different stakeholders across the life stages of projects. Often employing an experimental approach, these designers constantly move between deep research or scientific enquiry, and generative design thinking. To this end, the metabolic approach can be embedded across the life stages of projects, from change initiation and strategic planning (e.g. change of land use and accounting for associated emissions), to testing design alternatives (e.g. comparing low carbon configurations and choice of materials), all the way to use and maintenance (e.g. frequency of lawn mowing), recovery and end-of-life stages (e.g. decisions regarding how dead or felled trees should be managed). Key challenges include limited timeframes and the lack of allocated project budget for in-depth research and data collection on stock-flow-practices, and dealing with different actors with different disciplinary cultures including languages (Perrotti 2019).
4. *Design-led entrepreneurs* can have more impact on early stages of project initiation, planning and policy development stages. These designers often act as advocates for reforms in top-down governance systems, shaking up rigid frameworks and questioning business-as-usual approaches and regulations that often hinder resource-sensitive designs. The key leverage points include questioning tenders and project briefs for higher sustainability ambitions, negotiating the inclusion of performance metrics and assessment criteria and influencing policy using bottom-up approaches and facilitating design-led co-creation approaches, all important capabilities in boundary spanning (Nassauer 2023; van Dijk et al. 2023). Their agency increases through acquiring greater socio-political awareness of organizational and institutional arrangements, as well as entrepreneurial leadership skills while carrying design thinking capacities to push for creativity and innovation.

5.2. The contribution of design practice to urban metabolism studies

Despite a range of barriers identified in this study for employing metabolic thinking, our results also reveal important contributions of design to the broader discourse of UM. One key contribution is creating stronger links between often top-down resource accounting models, with on-the-ground spatial and social qualities, including everyday practices and lived experiences of places. Urban landscape designers possess a multilayered expertise in reading the site as palimpsests – a living organism and a continuum socio-ecological system within a meta-system of flows (Bobbink and de Wit 2020) – all critical to facilitating bottom-up metabolic approaches. With their training in both ecological and human sciences, and working closely with urban ecologists (Grose 2014) they engage with temporality, dynamic changes and fluxing ecologies in designing with living systems (Moosavi 2018), while accounting for the social and cultural particularities. Their 'design' skills enables them to describe and discover the needs and desires of people and places in a systemic way, and leverage design as '*a practice of cultural imagination*' (Hill 2022, 71). Carrying these skillsets can facilitate the integration of metabolic thinking, and even the adoption of UM approaches in projective and generative design processes and strategic design approaches when engaging with resources.

Additionally, the results show that 'stocks and flows' are perceived differently by designers, taking an expanded conceptualization. The term 'stock' is not explicitly used by designers, and 'flows' often take an expanded definition (e.g. fluxes of socio-ecological dynamics over time) (see Appendix, Table 5). The emphasis on socio-cultural 'practices' highlights the importance of understanding the relationships between practices, functions and services in UM studies. This resonates with emerging UM studies emphasizing a nexus approach that considers 'stocks-flows-services' or 'stocks-flows-practices' (Haberl et al. 2021). Urban landscape designers are uniquely positioned to bring an 'ecosystem services' lens to the understanding of the metabolism of cities, which then can contribute to a deeper understanding of flows dynamics and metabolic interdependencies in urban systems (Cárdenas-Mamani and Perrotti 2022; Perrotti and Stremke 2020). The examples of designers' efforts in shifting the clients' mindsets and everyday practices around resource use, exemplify the ways through which metabolic thinking in design can be impactful by looking beyond capitalist consumption, and considering metabolic interdependencies of design decisions.

Finally, the emphasis of the workshop participants on considering local climatic conditions and regulatory services provided by natural resources, highlights the importance of contextualizing and grounding metabolic approaches and understanding the existing environmental, temporal and climatic conditions that can greatly impact the stocks-flows-services relationships (Cárdenas-Mamani and Perrotti 2022).

While Landscape Architect Alan Berger in his book '*Systemic Design can change the world (2009)*' already highlighted landscape architects' role in improving material flows in cities, questions around 'agency' remain unexplored. This points to the need to understand how an enhanced agency in decisions related to resource stocks and flows can help cities' transition to a circular economy, and this research contributes to this important gap through empirical evidence.

5.3. Limitations of the study and research outlook

While the findings of the survey covered a wide geographical context, a potential bias exists due to greater response rates in certain countries where the authors have established networks. This limitation is typical in research using online surveys (Nayak and Narayan 2019). With regards to the focus group, the research can benefit from running multiple workshops in other countries with a larger number of participants to compare the findings with other contexts and address a potential contextual bias in this study. While the discourse of resource management and circularity is increasingly considered in the Belgian policy landscape, the notion of UM remains theoretical among design practitioners. Therefore, the ideas shared may not be representative of the broader landscape architecture community. Triangulation with the worldwide survey aimed at capturing these nuances.

Future research should include a deliberate protocol for shortlisting participants with a balance of different profiles. The quadrant matrix of designers' four different boundary spanning roles can be used to develop this protocol. The matrix can be further developed to include levers that can help designers increase their agency for resource-sensitive designs and offer pathways for playing multiple roles simultaneously in tackling complex socio-ecological challenges.

6. Conclusions

This study unpacks different modes of perceived agency of urban landscape designers and planners in embedding metabolic thinking in design/planning processes. It addresses an important gap which is the lack of studies that empirically investigate the understanding, perception and engagement of spatial practitioners with the concept of UM, through using a multi-method approach, and 'boundary spanning' as an analytical lens. Through this empirical inquiry, we conducted a worldwide survey with 101 spatial practitioners involved in urban landscape planning and design, a focus group workshop with 9 spatial practitioners mainly practicing in Belgium and 22 interviews with urban landscape designers across Belgium. The triangulation of data across the three methods allowed for an in-depth understanding of the ways designers use and understand the concept of UM, define metabolic flows, engage with data/tools to influence resource flows at different stages of design processes, and the perceived existing barriers.

We conclude that spatial practitioners often do not use UM to frame their understanding of resource flows and their metabolic interdependencies, and they rarely engage with quantification accounting models such as MFA or LCA, but they often incorporate metabolic thinking through a multi-scalar and iterative systems approach when working with living systems. Hence metabolic thinking can be conceptualized as incorporating systems thinking, and reading sites as multilayered palimpsests – socio-ecological systems in a constant state of becoming – thus integrating social, territorial and temporal dimensions in understanding resource flows beyond the sites of intervention.

The findings reveal key contributions of urban landscape architects to UM studies. This includes offering an expanded understanding of the stocks-flows-practices nexus (Haberl et al. 2021). Having a wide range of expertise across environmental science, humanities and spatial systems, urban landscape designers can play a critical role in connecting metabolic stocks and flows to socio-cultural practices, while considering the spatial–temporal dimensions often overlooked in UM studies. This expands UM approaches beyond standardized input-output analytical methods, towards a greater engagement with socio-cultural practices that define and drive resource flows. These findings further advance the associated projective and generative potential of metabolic thinking in design by offering alternative visions for the transformation of urban environments.

Embedding metabolic thinking in design processes comes with multiple challenges. Existing knowledge and information barriers, projects' budgetary and time restrictions, limited data and tools adapted to landscape design, clients' values and mindsets, and rigid project briefs and inflexible policies that enable resource circularity were key barriers identified by designers who participated in this study.

This study provides important implications for planning and policy, by emphasizing the role of landscape-led and nature-based approaches in enhancing the metabolism of cities. Greater socio-political engagement of designers to inform (and reform) policies and regulations at early stages of projects can advance their agency in transforming practices towards resource-sensitive and climate responsive designs. For example, the *Circular Economy Action Plan* (European Commission 2020) promotes initiatives to reduce soil sealing, rehabilitate abandoned or contaminated brownfields and increase the safe, sustainable and circular use of excavated soils, which require design knowledge and technical inputs from landscape architects at early stages of projects.

To transform the ways our cities are designed and built in relation to natural systems, there is a need for alternative design strategies and methods. To this end, urban landscape designers can provide important contributions by bringing natural processes to the fore, but they need to cross multiple boundaries and voice their agency in improving the metabolism of cities. This means acquiring new boundary spanning skills and improving their coordination, negotiation and communication skills to lever transdisciplinary approaches in linking science-policy-practice. They also need to think beyond the constructed physical, social and disciplinary boundaries, and continue developing and reinventing the roles they can play in orchestrating sustainable resource management. This may entail reforms in education systems in built environment disciplines, to include boundary spanning competencies, a key topic for future research.

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